

Comparative Risk of Hyperglycemia With PI3K, AKT, and mTOR Inhibitors in Breast Cancer: A Systematic Review and Meta-Analysis

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Purpose

To compare the risk of hyperglycemia associated with PI3K, AKT, and mTOR inhibitors in patients with breast cancer and to characterize differences in risk across drug classes and severity grades.

Background

Targeting the PI3K-AKT-mTOR (PAM) signaling pathway has become a key therapeutic strategy in advanced breast cancer, particularly in hormone receptor-positive disease. Agents across this pathway, including PI3K inhibitors, AKT inhibitors, and mTOR inhibitors, improve disease control but can disrupt insulin signaling, leading to treatment-related hyperglycemia. Despite widespread clinical use, the comparative risk of hyperglycemia across these agents and classes remains incompletely defined, limiting optimal risk stratification and management.

Methods

A systematic literature search was conducted across PubMed/MEDLINE, Embase (OVID), CENTRAL, ClinicalTrials.gov, and Web of Science from database inception through January 1, 2025. Randomized controlled trials in breast cancer comparing a PAM pathway inhibitor at standard dosing with a control regimen lacking that agent were included. Standard doses analyzed included alpelisib 300 mg, inavolisib 9 mg, taselisib 4 mg, capivasertib 400 mg twice daily, and everolimus 10 mg. Trials reporting any-grade hyperglycemia were eligible. Risk ratios (RRs) for any-grade and grade ≥ 3 hyperglycemia were extracted and synthesized.

Results

Fourteen randomized trials including 6,940 patients were analyzed. All PAM pathway inhibitors were associated with increased hyperglycemia risk compared with control regimens. PI3K inhibitors demonstrated the highest risk, particularly alpelisib (any-grade RR 6.90; 95% CI, 5.01-9.51; grade ≥ 3 RR 58.75; 95% CI, 16.96-203.49). Inavolisib also showed increased risk (any-grade RR 4.69; 95% CI, 3.13-7.04; grade ≥ 3 RR 23.28; 95% CI, 1.38-391.84). AKT inhibitors and taselisib demonstrated intermediate risk (capivasertib: any-grade RR 3.53; 95% CI, 2.42-5.16; grade ≥ 3 RR 10.02; 95% CI, 2.70-37.16; taselisib: any-grade RR 3.19; 95% CI, 1.64-6.21; grade ≥ 3 RR 14.61; 95% CI, 2.79-76.38). mTOR inhibition with everolimus demonstrated the lowest relative risk (any-grade RR 2.19; 95% CI, 1.80-2.68), with no statistically significant increase in grade ≥ 3 hyperglycemia (RR 5.89; 95% CI, 0.74-47.13).

Implications

All PAM pathway inhibitors increase the risk of hyperglycemia, with a clear gradient across drug classes. PI3K inhibitors carry the highest risk, followed by AKT inhibitors, with mTOR inhibitors demonstrating comparatively lower risk. This gradient likely reflects the proximity of pathway inhibition to insulin receptor signaling. These findings support proactive metabolic monitoring and risk stratification, particularly for patients receiving PI3K inhibitors, and may inform agent selection in patients with baseline metabolic risk. Further studies are needed to refine management strategies and evaluate newer agents.

